



Comparison Of Arima And Arimax Model Forecasts Of Monthly Sales Of Four Petroleum Products In Nigeria

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ABSTRACT

This research was carried out using statistical software's as Minitab 18, Grelt and Micro-soft Excel. Monthly sales of petroleum products in Nigeria National Petroleum Corporation [NNPC] from 2007 to 2018 such as PMS, HNK ATK and AGO were collected. The monthly and yearly means plots were done, so as to have a better understanding of the series behaviors. Then, stationarity of the sales of petroleum products were obtained by first difference since a slightly linear exists of all the series. A suitable ARIMA and ARIMAX model for each of the four variables were identified. Finally, the fitted and forecast values of the identified ARIMA and ARIMAX model were obtained and compared, using Forecast Accuracy Measures. The results of the identified ARIMAX models forecasts (2019 to 2020) for each variables shows appreciate as the years goes for three variables (PMS, HNK and AGO) except ATK variable, where the plot will depreciate. In addition, the identified ARIMA models forecasts only appreciate for ATK variable while it's depreciated for PMS, HNK and AGO variables. Hence, the identified ARIMAX models performance better than the ARIMA models for three of the variables (PMS, HNK and AGO), using accuracy measure (MAE, MSE and RMSE) except ATK. This research recommended ARIMAX models for the forecasting PMS, HNK and AGO for effective and reliable understanding of this variables, except ATK variable. Also, recommended ARIMA model for ATK variable.

Keywords: ARIMA Model, ARIMAX Model, Accuracy Measure, Stationarity, Fitted and Forecast values

1.0 INTRODUCTION

Crude oil is a vital commodity in the world market despite the crusade for green energy and other sources of power. Crude oil is still an expensive commodity in the international market. Crude oil which is the most dominant source of energy in Nigeria was discovered pre to 1956. The univariate time series models, including the autoregressive integrated moving average (ARIMA) model and its derivatives such as Seasonal ARIMA (SARIMA), periodic ARIMA, and de-seasonalized ARIMA models, have long been applied in stream flow forecasting, particularly in the modeling of monthly stream flow (1-6). Time series forecasting is an important area of forecasting in which past observations of the same variable are collected and analyzed to develop a model describing the underlying relationship. The model is then used

to extrapolate the time series into the future. This modeling approach is particularly useful when little knowledge is available on the underlying data generating process or when there is no satisfactory explanatory model that relates the prediction variable to other explanatory variables. Much effort has been devoted over the past several decades to the development and improvement of time series forecasting models.

One of the most important and widely used time series models is the autoregressive integrated moving average (ARIMA) model. The popularity of the ARIMA model is due to its statistical properties as well as the well-known Box-Jenkins methodology in the model building process. Nigeria was the second oil producing nation in Africa after Libya and sixth in the world (7). Now, Nigeria is the largest oil producing country in Africa -. Here, the oil industry generates around 2.25 million barrels of oil on a daily basis, which represents approximately 113 million tonnes every year. Since the middle of the 20th century, the country has been developing its oil processing industry. Today, oil makes up 90 percent of total exports from Nigeria and generates around 80 percent of its revenue according to worlds atlas.com/articles/top 10 oil producing countries in Africa.html. This study was prompted by the irregularity in the sales of petroleum products in Nigerian market. Over the years, the act of regulating and reviewing the prices of petroleum products including PMS, AGO, ATK and HHK etc. has gained a space in the hearts of Nigeria and most of the time the out of stock of petroleum products. In view of the above statement, this research seeks to examine the comparison of ARIMA and ARIMAX models forecasts of monthly sales of these four petroleum products in Nigeria. This research seeks to examine monthly sales of petroleum products in Nigeria National Petroleum Corporation [NNPC] from 2007 to 2018. To achieve our aim, the set of objectives are;

1. To determine the trend components and stationarity of sales of petroleum products.
2. To build a suitable ARIMA and ARIMAX model for the four variables.
3. Obtain suitable forecasts for suitable ARIMA and ARIMAX models (four variables).
4. To compare the fitted and forecast values of ARIMA and ARIMAX model using Forecast Accuracy Measures [FAM].

This study of monthly sales of petroleum products in time series is limited to the data obtained from Nigeria National Petroleum Corporation [NNPC] from January to December 2007-2018. The data covers only Premium Motor Spirit [PMS], House Hold Kerosene [HHK], Automotive Gas Oil [AGO] and Aviation Turbine Kerosene [ATK]. This study will cover only ARIMA and ARIMAX of monthly sales of four petroleum products in Nigeria. The outcome of this research works would be beneficial to students, statistician and others as a reference to other prediction to future researches that will be done in related area.

ARIMA modeling techniques have been applied in many fields of research. One of the primary benefits of ARIMA models is their ability to correct for local trends in the data – what has happened on the previous day is incorporated into the forecast of what will happen today (8). This works well, for example, during a particularly severe flu season, where prolonged periods of high visit rates are adjusted to by the ARIMA model, thus preventing the alarm from being triggered every day throughout the flu season. However, if the ARIMA model “adjusts” to an actual outbreak instead of detecting it, a slowly spreading outbreak or attack might be missed because of this correction. This correction is most likely to affect detection of outbreaks occurring over several days, rather than those that occur suddenly. It is therefore also important to rely on then on-ARIMA or non-classical ARIMA models for outbreak detection.

3.0 METHODOLOGY

3.1 Data

The data used in the study was gotten from the official website of the Nigeria National Petroleum Operation (NNPC) covering a period from January 2007 to December 2018. The data comprises of eleven

years with 144 observations. The statistical tool (package) used to analyze the data of this study are MINITAB, EXCEL and GRET. L.

3.2 Materials

The methodology adopted for this study is the general time series methodology introduced by Box and Jenkins (9). Box and Jenkins, applied ARIMA models to find the best fit of a time series data in order to make forecasts. The model is generally referred to as an ARIMA (p, d, q) model where p, d and q are non-negative integers that refer to the order of the autoregressive, integrated, and moving average parts of the model respectively. The Box-Jenkins model uses three stage iterative modeling approach which are:

1. **Model identification and model selection:** Making sure that the variables are stationary, identifying seasonality in the dependent series (seasonally differencing it if necessary), and using plots of the autocorrelation and partial autocorrelation functions of the dependent time series to decide which (if any) autoregressive or moving average component should be used in the model. **Stationarity and Seasonality.** The first step in developing a Box-Jenkins model is to determine if the time series is stationary and if there is any significant seasonality that needs to be modeled. Stationarity can be assessed from a run sequence plot. The run sequence plot should show constant location and scale. It can also be detected from an autocorrelation plot. Specifically, non-stationarity is often indicated by an autocorrelation plot with very slow decay. Finally, unit root tests provide a more formal approach to determining the degree of differencing such as Kwiatkowski-Phillips-Schmidt-Shin (KPSS) and Phillips-Perron Unit Root Tests. The KPSS test for the null hypothesis of a level stationary against an alternative of unit root together with the Philips-Peron test for the null hypothesis of a unit root against the alternative of a stationary series. (10) The decision rule is that for the KPSS test if the p -value of its test statistic is greater than the critical value, say 0.05, then reject the null hypothesis of having a level stationary series and therefore conclude the alternate hypothesis that it has a unit root. The Philips Peron Test, on the other hand, test for the null hypothesis of unit root against an alternative hypothesis of stationarity by rejecting the null hypothesis if its p -value is less than the critical value chosen.

Differencing to achieve stationarity: Box and Jenkins recommended the differencing approach to achieve stationarity. However, fitting a curve and subtracting the fitted values from the original data can also be used in the context of Box-Jenkins models.

Seasonal Differencing: At the model identification stage, the goal is to detect seasonality, if it exists, and to identify the order for the seasonal autoregressive and seasonal moving average terms. For many series, the period is known and a single seasonality term is sufficient. For example, for monthly data one would typically include either a seasonal AR 12 term or a seasonal MA 12 term. For Box-Jenkins models, one does not explicitly remove seasonality before fitting the model. Instead, one includes the order of the seasonal terms in the model specification to the ARIMA estimation software. However, it may be helpful to apply a seasonal difference to the data and regenerate the autocorrelation and partial autocorrelation plots. This may help in the model identification of the non-seasonal component of the model. In some cases, the seasonal differencing may remove most or all of the seasonality effect.

Identifying p and q

Once stationarity and seasonality have been addressed, the next step is to identify the order (i.e. the p and q) of the autoregressive and moving average terms. These are determined by examining the values of the autocorrelations and the partial autocorrelations with their corresponding plots as explained below.

Autocorrelation and Partial Autocorrelation Plots

The primary tools for doing this are the autocorrelation plot and the partial autocorrelation plot. The sample autocorrelation plot and the sample partial autocorrelation plot are compared to the theoretical behavior of these plots when the order is known.

2. Parameter Estimation

Parameter estimation uses computation algorithms to arrive at coefficients which best fit the selected ARIMA model. The most common methods use maximum likelihood estimation or non-linear least-squares estimation.

3. Model checking

Model checking is done by testing whether the estimated model conforms to the specifications of a stationary univariate process. In particular, the residuals should be independent of each other and constant in mean and variance over time (plotting the mean and variance of residuals over time and performing a Ljung-Box test or plotting autocorrelation and partial autocorrelation of the residuals are helpful to identify misspecification). If the estimation is inadequate, we have to return to step one and attempt to build a better model.

This procedure includes the following steps.

4.0 RESULT

The data comprises of eleven years with 144 observations. In the section, compare of the identified ARIMA and ARIMAX model fitted and forecast values were done.

4.1 Results for PMS analysis

Figure 1 shows that the series depreciate at the beginning, then appreciate as the years goes. It is an indication that the series is not stationary. In addition, the ACF plot suggest an AR(p) process and PACF suggest AR (2) since the first two lags cut on. Then, the series was difference Fig. 3 below showed that the series is stationary and the ACF plot suggest an AR(p) process and PACF suggest AR (1) since only the first lag cut on.

Fig1.0 Actual SERIES Time Plots for PMS

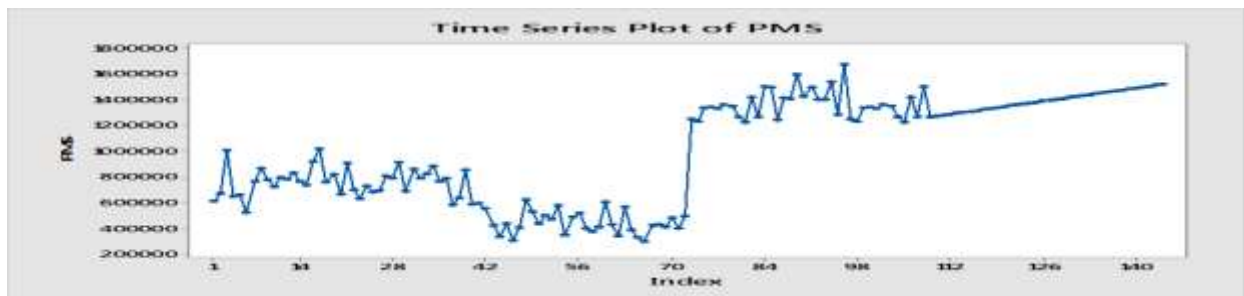


Fig2.0 Actual Series ACF and PACF Plots for PMS

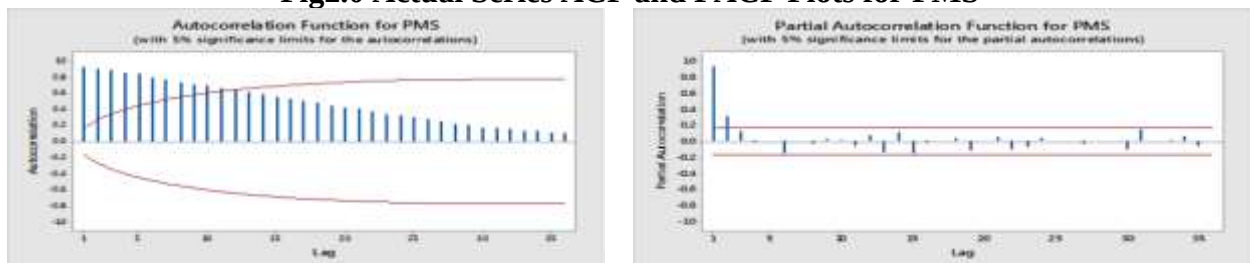


Fig 3.0 Series Plots for 1st Difference for PMS

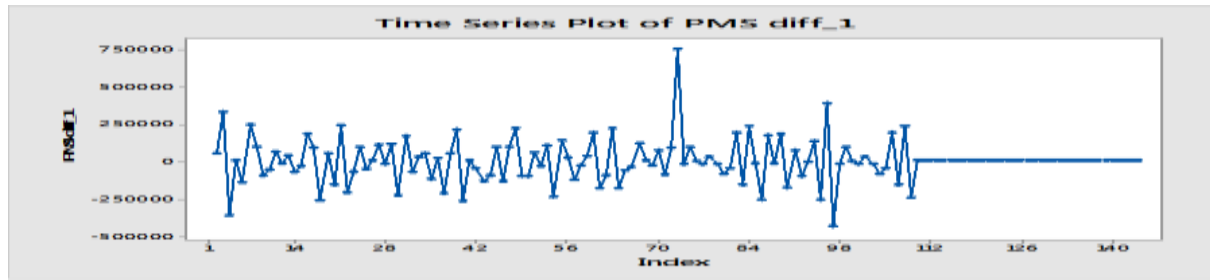
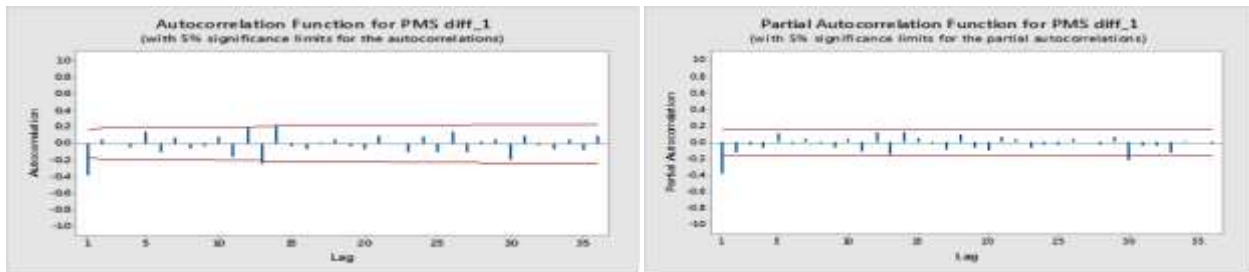


Fig4.0 ACF and PACF Plots for 1st Difference for PMS



ARIMA Model Building and Its Parameter Estimates

Various ARIMA models were fitted to the data sets PMS using the ACF and PACF correlograms for the actual and first difference series, which suggested AR (1). The results are summarized below in Table 1

Table 1: ARIMA Models, Parameter Estimates and Model Selection Criteria for PMS

Models	ϕ_1	ϕ_2	Θ_1	Θ_2	RSS	K	AIC	BIC
ARIMA (1,1,0)	-0.3828 (0.000)				17055178197	1	3392.586	4111.209
ARIMA (2,1,0)	-0.4331 (0.000)	-0.1303 (0.121)			16874761641	2	3392.037	4113.629
ARIMA (0,1,1)			0.4107 (0.000)		16802556906	1	2727.292	3445.915
ARIMA (0,1,2)			0.4378 (0.000)	-0.0554 (0.511)	16866443832	2	3391.966	4113.558
ARIMA (1,1,1)	-0.1220 (0.042)		0.3131 (0.103)		16873240100	1	2677.402	3396.025
ARIMA (1,1,2)	-0.7666 (0.970)		-0.3562 (0.986)	0.3153 (0.970)	17042469220	2	Correlated Residual	Correlated Residual
ARIMA (2,1,1)	-1.0353 (0.005)	-0.3416 (0.009)	-0.6210 (0.101)		16964805569	2	2680.181	3401.774
ARIMA (2,1,2)	0.5865 (0.503)	-0.0639 (0.801)	1.0262 (0.240)	-0.3533 (0.230)	17039588051	2	Correlated Residual	Correlated Residual

Footnote: the bold model is the best model, because it has the least AIC and BIC

The identified Models are ARIMA (1,1,1)

$$\text{ARIMA (1,1,1): } (1 - \phi_1 B) \nabla X_t = e_t + \theta_1 e_{t-1} \quad \text{eq.1}$$

By substitution

$$\text{ARIMA (1,1,1): } (1 + 0.12B) \nabla X_t = e_t + 0.31e_{t-1} \quad \text{eq.2}$$

4.2 Results for HNK analysis

Figure 3 showed a slightly depreciate and an upward appreciate as the years goes. It is an indication that the series is not stationary. In addition, the ACF plot suggest an AR(p) process in suggest AR (2) since the first two lags cut. Then, the series was differenced ones to obtain stationarity (first-difference). Figure 4. shows that the series is stationary and the ACF plot suggest an AR (1) process and PACF suggest AR (1) since only the first lag cuts.

Fig 5.0 Actual Series Time for HNK

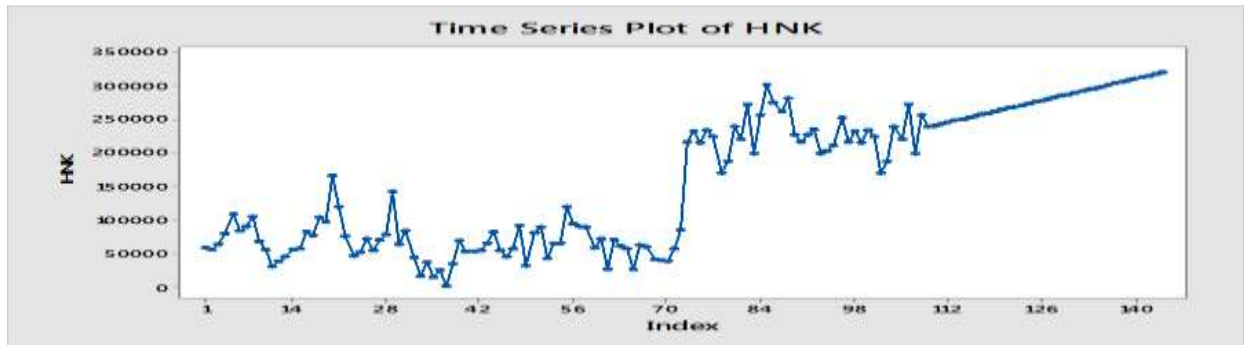


Fig 6.0 Actual Series ACF and PACF Plots for HNK

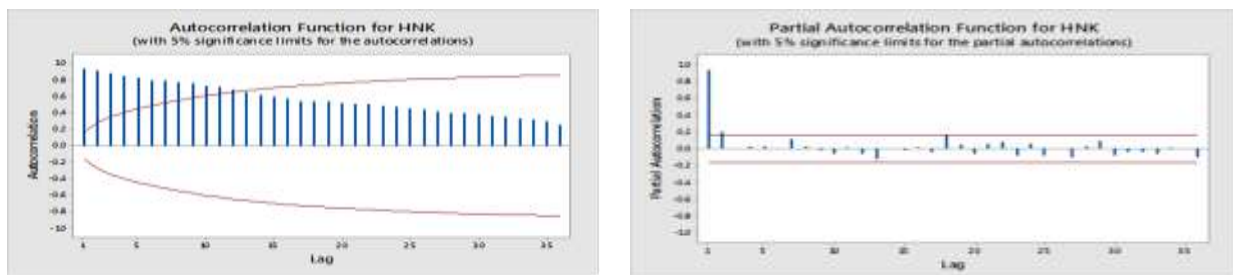
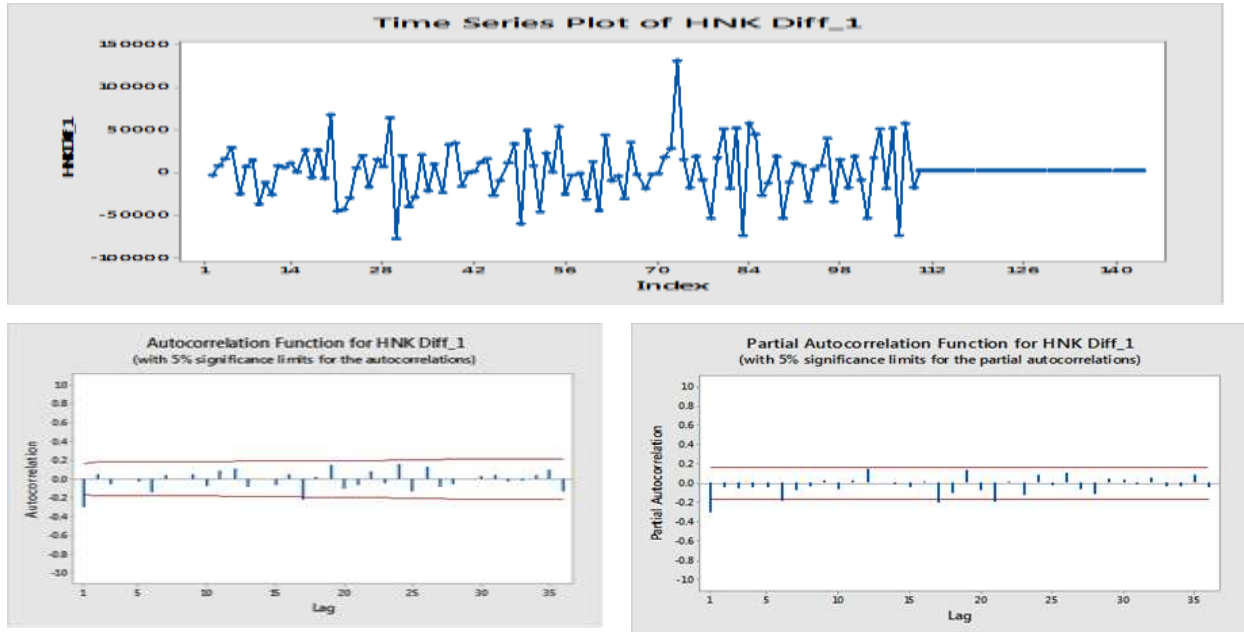


Fig7.0 Series, ACF and PACF Plots for 1st Difference for HNK



ARIMA Model Building and Its Parameter Estimates

Various ARIMA models were fitted to the data sets HNK using the ACF and PACF correlograms for the actual and first difference series. The results are summarized below in Table2

Table 2: ARIMA Models, Parameter Estimates and Model Selection Criteria for HNK

Models	ϕ_1	ϕ_2	θ_1	θ_2	RSS	K	AIC	BIC
ARIMA (1,1,0)	-0.2952 (0.000)				787338345	1	2949.706	3668.329
ARIMA (2,1,0)	-0.3038 (0.000)	-0.0291 (0.730)			792252563	2	2951.585	3673.177
ARIMA (0,1,1)			0.3021 (0.000)		787315860	1	2618.13	3336.753
ARIMA (0,1,2)			0.3051 (0.000)	-0.0278 (0.741)	792387121	2	2951.609	3673.202
ARIMA (1,1,1)	-0.1587 (0.568)		0.1486 (0.593)		791927351	1	Correlated Residual	Correlated Residual
ARIMA (1,1,2)	0.7848 (0.000)		1.0858 (0.000)	-0.2002 (0.051)	793462226	2	2239.183	2960.776
ARIMA (2,1,1)	0.5245 (0.014)	0.1875 (0.124)	0.8476 (0.000)		7888712226	2	2569.92	3291.512
ARIMA (2,1,2)	0.3170 (0.605)	0.2868 (0.285)	0.6352 (0.307)	0.1539 (0.700)	793524001	2	Correlated Residual	Correlated Residual

Footnote: the bold model is the best model, because it has the least AIC and BIC

$$\text{ARIMA}(1,1,2): (1 - \phi_1 B) \nabla X_t = e_t + \theta_1 e_{t-1} + \theta_2 e_{t-2} \quad \text{eq.3}$$

By substitution

$$\text{ARIMA}(1,1,2): (1 - 0.78B) \nabla X_t = e_t + 1.09e_{t-1} - 0.20e_{t-2} \quad \text{eq.4}$$

4.3 Results for ATK analysis

Figure 3.0 shows an upward and downward movement as the years goes. It is an indication that the series is not stationary. In addition, the ACF plot suggest an AR (3) process and PACF suggest AR (2) since the first two lags cut. Then, the series was differenced once to obtain stationarity (first-difference). Figure 9 shows that the series is stationary and the ACF plot suggest an AR (1) process and PACF suggest AR (1) since only the first lag cut on Figure8, and 9.

Fig8.0 Actual SERIES, ACF and PACF Plots for ATK

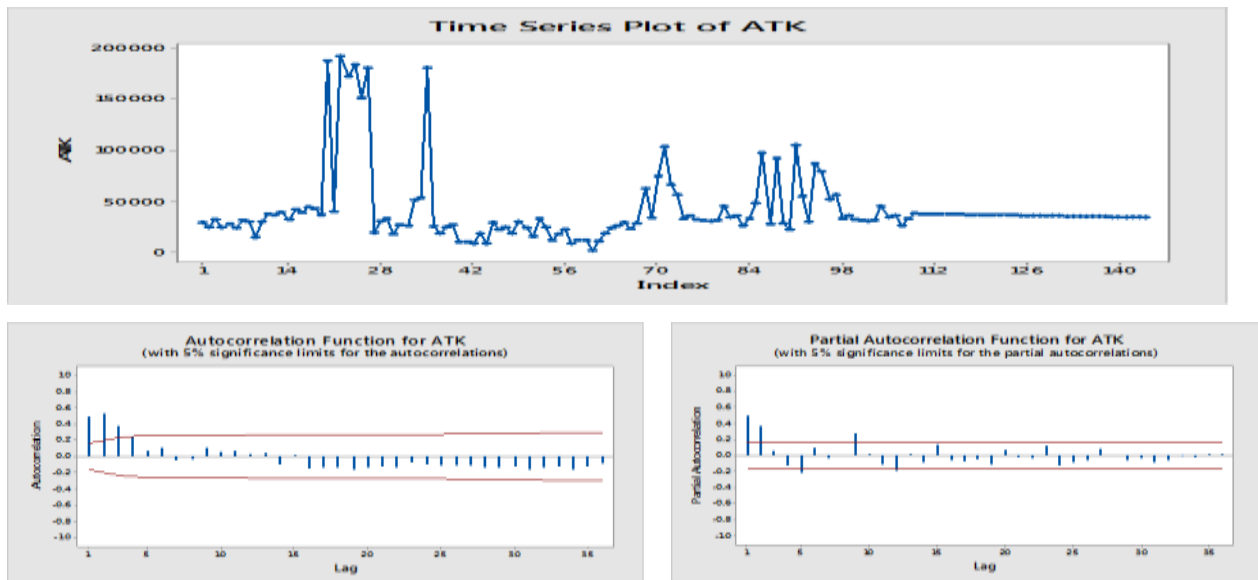
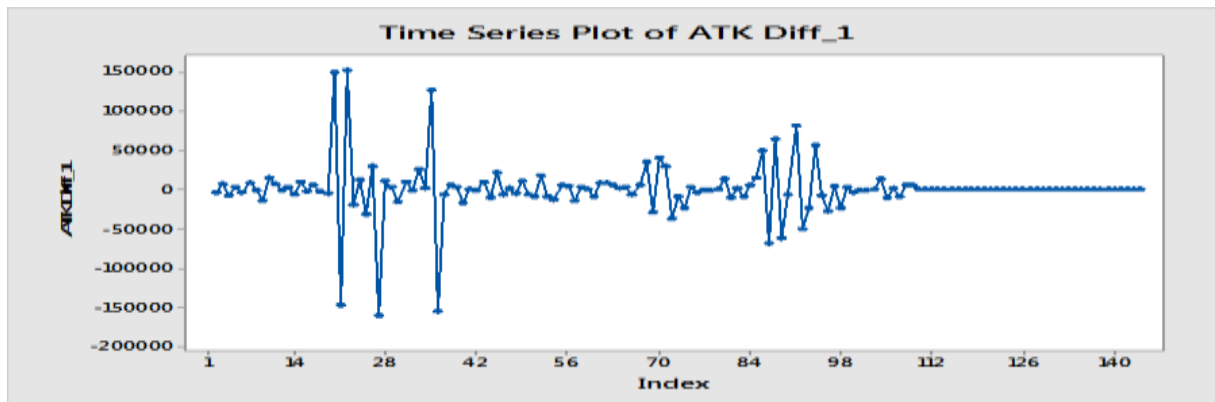
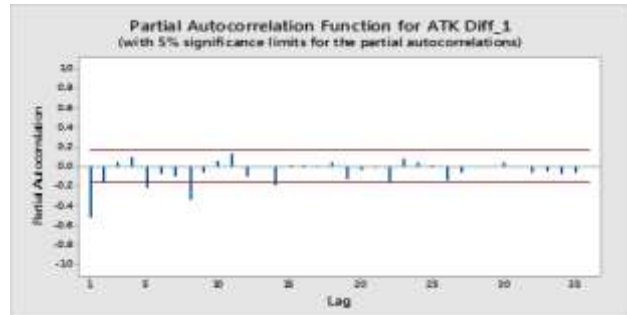
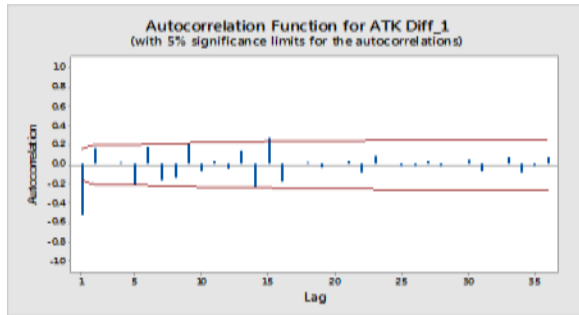


Fig9.0 Series, ACF and PACF Plots for 1st Difference for ATK





ARIMA Model Building and Its Parameter Estimates

Various ARIMA models were fitted to the data sets ATK using the ACF and PACF correlograms for the actual and first difference series. The results are summarized below in Table3

Table3 ARIMA Models, Parameter Estimates and Model Selection Criteria for ATK

Models	ϕ_1	ϕ_2	Θ_1	Θ_2	Θ_3	RSS	K	AIC	BIC
ARIMA (1,1,0)	-0.5292 (0.000)					901320352	1	2637.6	3356.223
ARIMA (2,1,0)	-0.0.6085 (0.000)	-0.1500 (0.074)				887305555	2	2967.901	3689.494
ARIMA (0,1,1)			0.5395 (0.000)			916150485	1	2971.526	3690.148
ARIMA (0,1,2)			0.5855 (0.000)	-0.2776 (0.001)		885456535	2	2967.601	3689.193
ARIMA (1,1,1)	-0.3709 (0.010)		0.2274 (0.128)			890929284	1	2244.578	2972.49
ARIMA (1,1,2)	-0.1059 (0.736)		0.4904 (0.112)	-0.2203 (0.186)		890858898	2	Correlated Residual	Correlated Residual
ARIMA (2,1,1)	0.3186 (0.000)	0.3769 (0.000)	0.9993 (0.000)			823752670	2	2244.578	2966.17
ARIMA (2,1,2)	0.5153 (0.000)	0.2664 (0.002)	1.2204 (0.000)	-0.2213 (0.000)		832730828	2	2246.139	2967.731
ARIMA (1,1,3)	-0.7455 (0.611)		-0.1556 (0.916)	0.1552 (0.861)	- 0.1842 (0.679)	897635240	3	Correlated Residual	Correlated Residual
ARIMA (0,1,3)			0.6549 (0.000)	-0.2915 (0.003)	0.1529 (0.069)	886791980	3	2257.196	2981.759

Footnote: the bold model is the best model, because it has the least AIC and BIC

$$\text{ARIMA}(2,1,1): (1 - \phi_1 B - \phi_2 B^2) \nabla X_t = e_t + \theta_1 e_{t-1} \quad \text{eq.5}$$

By substitution

$$\text{ARIMA}(2,1,1): (1 - 0.319B - 0.377B^2) \nabla X_t = e_t + 0.999e_{t-1} \quad \text{eq.6}$$

Results for AGO analysis

Figure 10 shows a slightly depreciate and an upward appreciate as the years goes. It is an indication that the series is not stationary. In addition, the ACF plot suggest an AR(p) process in Figure 10 PACF suggest AR (2) since the first two lags cut in Figure 10. Then, the series was difference once to obtain stationarity (first-difference) in Figure11. Figure 12 shows that the series is stationary and the ACF plot suggest an AR (1) process and PACF suggest AR (1) since only the first lag cut on Figure 11 and 12.

Fig 10.0 Actual SERIES, ACF and PACF Plots for AGO

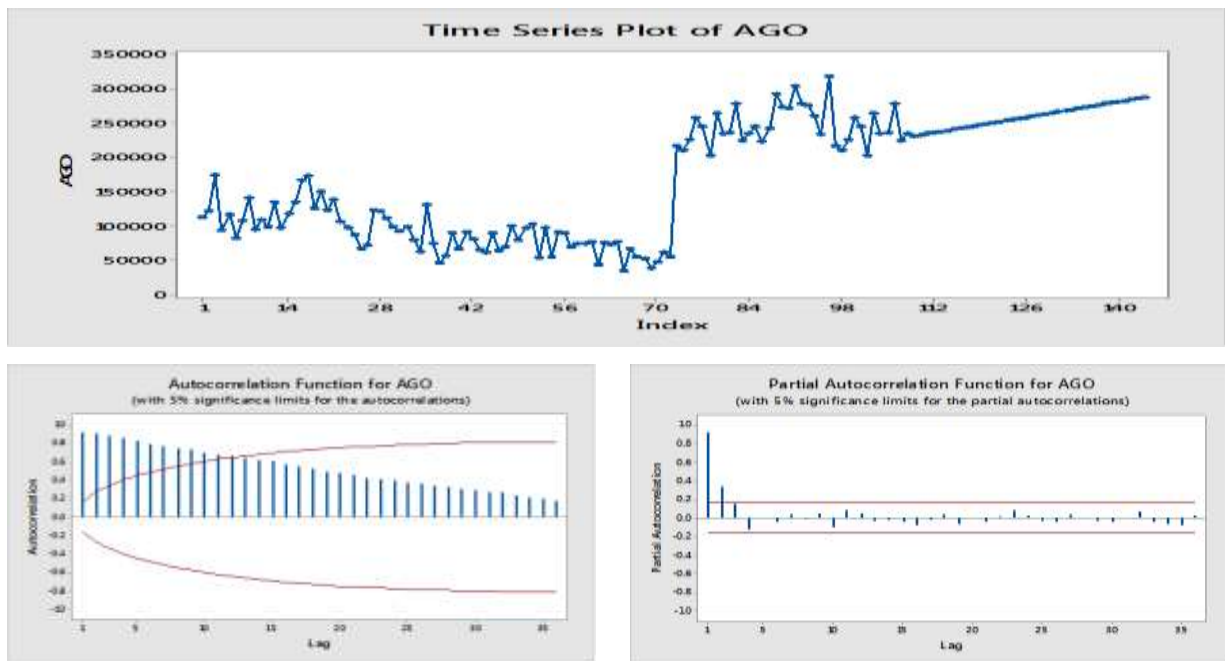


Fig11.0 Time Plots for 1st Difference for AGO

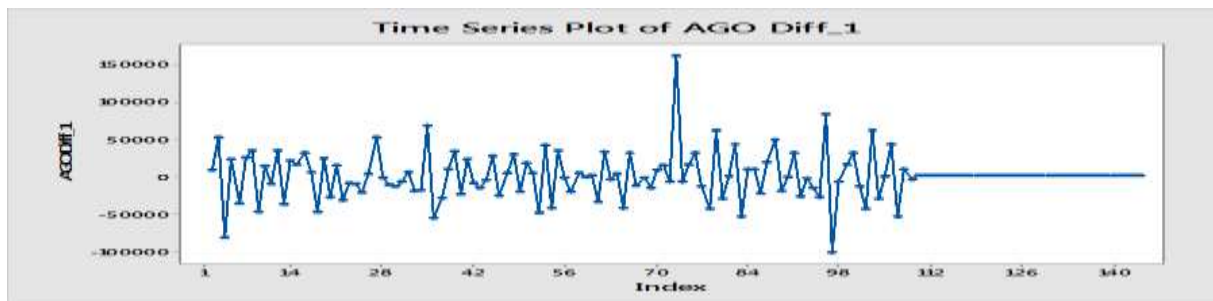
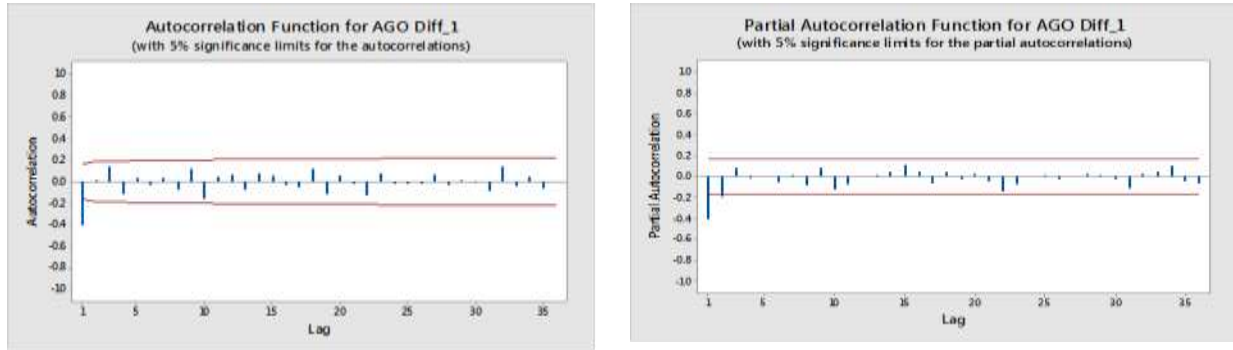


Fig12.0 ACF and PACF Plots for 1st Difference for AGO



ARIMA Model Building and Its Parameter Estimates

Various ARIMA models were fitted to the data sets AGO using the ACF and PACF correlograms for the actual and first difference series. The results are summarized below in Table

Table 4.: ARIMA Models, Parameter Estimates and Model Selection Criteria for AGO

Models	ϕ_1	ϕ_2	Θ_1	Θ_2	RSS	K	AIC	BIC
ARIMA (1,1,0)	-0.4094 (0.000)				778799451	1	2610.318	3328.941
ARIMA (2,1,0)	-0.4908 (0.000)	-0.1968 (0.018)			753649650	2	2944.391	3665.984
ARIMA (0,1,1)			0.4318 (0.000)		765402442	1	2945.637	3664.26
ARIMA (0,1,2)			0.4920 (0.000)	-0.1297 (0.123)	760884176	2	2945.767	3667.36
ARIMA (1,1,1)	-0.1790 (0.332)		0.2950 (0.101)		764753043	1	Correlated Residual	Correlated Residual
ARIMA (1,1,2)	-0.8561 (0.000)		-0.4527 (0.029)	0.4076 (0.000)	769803192	2	2234.824	2956.417
ARIMA (2,1,1)	0.8389 (0.000)	0.1592 (0.048)	1.0100 (0.000)		837931013	2	2247.035	2968.628
ARIMA (2,1,2)	-1.1197 (0.000)	-0.3540 (0.039)	-0.6591 (0.001)	0.1185 (0.532)	751935706	2	2231.442	2953.035

Footnote: the bold model is the best model, because it has the least AIC and BIC

ARIMA (2,1,2) for AGO.

$$\text{ARIMA (2,1,2): } (1 - \phi_1 B - \phi_2 B^2) \nabla X_t = e_t + \theta_1 e_{t-1} + \theta_2 e_{t-2} \quad \text{eq.7}$$

By substitution

$$\text{ARIMA (2,1,2): } (1 + 1.11B + 0.35B^2) \nabla X_t = e_t - 0.66e_{t-1} + 0.12e_{t-2} \quad \text{eq.8}$$

5.0 ARIMAX Model Building and Its Parameter Estimates

In this section, several ARIMAX model were built to each variable with the others as the independent variables. Then, the model selection criteria AIC and BIC were used to identified the best model among the models. The specify model are as follow;

$$\text{Model A} \quad Y_t = \emptyset_1 Y_{t-1} + \beta_1 X_t + \beta_2 Z_t + \beta_3 W_t + e_t \quad \text{eq.9}$$

$$\text{Model B} \quad Y_t = \emptyset_1 e_{t-1} + \beta_1 X_t + \beta_2 Z_t + \beta_3 W_t + e_t \quad \text{eq.10}$$

$$\text{Model C} \quad Y_t = \emptyset_1 Y_{t-1} + \emptyset_2 Y_{t-2} + \beta_1 X_t + \beta_2 Z_t + \beta_3 W_t + e_t \quad \text{eq.11}$$

$$\text{Model D} \quad Y_t = \emptyset_1 e_{t-1} + \emptyset_2 e_{t-2} + \beta_1 X_t + \beta_2 Z_t + \beta_3 W_t + e_t \quad \text{eq.12}$$

$$\text{Model E} \quad Y_t = \emptyset_1 Y_{t-1} + \emptyset_1 e_{t-1} + \beta_1 X_t + \beta_2 Z_t + \beta_3 W_t + e_t \quad \text{eq.13}$$

$$\text{Model F} \quad \nabla Y_t = \emptyset_1 Y_{t-1} + \beta_1 X_t + \beta_2 Z_t + \beta_3 W_t + e_t \quad \text{eq.14}$$

$$\text{Model G} \quad \nabla Y_t = \emptyset_1 e_{t-1} + \beta_1 X_t + \beta_2 Z_t + \beta_3 W_t + e_t \quad \text{eq.15}$$

$$\text{Model H} \quad \nabla Y_t = \emptyset_1 Y_{t-1} + \emptyset_1 e_{t-1} + \beta_1 X_t + \beta_2 Z_t + \beta_3 W_t + e_t \quad \text{eq.16}$$

where,

Y_t is PMS variable, X_t is HNK variable, Z_t is ATK variable, W_t is AGO variable and ∇ is first difference.

Table 5: ARIMAX Models, Parameter Estimates and Model Selection Criteria for PMS

MODELS	Φ_1	Φ_2	Θ_1	Θ_2	β_1	B ₂	B ₃	AIC	BIC
ARIMAX MODEL A (PMS)	0.974964 (0.0000)				0.694892 (0.0398)	- 0.806760 (0.0007)	2.84906 (0.00001)	3735.718	3741.752
ARIMAX MODEL B (PMS)			0.271277 (0.00009)		-0.279058 (0.4384)	0.911024 (0.0037)	5.64548 (0.00001)	3791.861	3797.894
ARIMAX MODEL C (PMS)	0.560424 (0.00001)	0.425113 (0.00001)			0.282461 (0.38690)	- 0.652582 (0.01037)	3.17245 (0.00001)	3711.022	3718.263
ARIMAX MODEL D (PMS)			0.296293 (0.0030)	0.323719 (0.0002)	0.0711220 (0.8615)	0.614132 (0.0768)	5.32642 (0.0001)	3780.052	3787.293
ARIMAX MODEL E (PMS)	- 0.432028 (0.00001)				0.246279 (0.4473)	- 0.660922 (0.0092)	3.13532 (0.00001)	3680.875	3686.895)
ARIMAX MODEL F (PMS)	- 0.371537 (0.00001)				-0.141358 (0.5769)	- 0.196346 (0.3895)	0.225096 (0.3967)	3781.684	3787.704
ARIMAX MODEL G (PMS)			- 0.405889 (0.00001)		- 0.0316238 (0.8860)	0.106929 (0.6014)	0.0950943 (0.6863)	3780.103	3786.123
ARIMAX MODEL H (PMS)	- 0.123767 (0.5156)		- 0.303799 (0.1028)		-0.118265 (0.8460)	- 0.118265 (0.5723)	0.112800 (0.6571)	3781.684	3788.907

Footnote: the bold model is the best model, because it has the least AIC and BIC

Therefore, the identified ARIMAX model in Table 5 is

$$\text{Model (E): } Y_t = \phi_1 Y_{t-1} + \theta_1 e_{t-1} + \beta_1 X_t + \beta_2 Z_t + \beta_3 W_t + e_t \quad \text{eq.17}$$

By substitution

$$Y_t = 0.43Y_{t-1} + 0.24e_{t-1} + 0.24X_t - 0.66Z_t + 3.14W_t + e_t \quad \text{eq.18}$$

Table 6: ARIMAX Models, Parameter Estimates and Model Selection Criteria for HNK

MODELS	Φ_1	Φ_2	Θ_1	Θ_2	β_1	B_2	B_3	AIC	BIC
ARIMAX MODEL A (HNK)	0.906019 (0.00001)				0.0851639 (0.18803)	0.405024 (0.00021)	0.0594412 (0.03343)	3331.760	3337.794
ARIMAX MODEL B (HNK)			0.421563 (0.00001)		-0.104171 (0.15844)	0.965423 (0.00001)	0.0012169 (0.95617)	3389.625	3395.659
ARIMAX MODEL C (HNK)	0.668834 (0.00001)	0.293586 (0.00045)			0.0410836 (0.5427)	0.46229 (0.00001)	0.0317999 (0.17905)	3322.209	3329.449
ARIMAX MODEL D (HNK)			0.60213 (0.00001)	0.454412 (0.00001)	- 0.0106906 (0.8794)	0.730023 (0.00001)	0.0376647 (0.0656)	3360.100	3367.340
ARIMAX MODEL E (HNK)	0.979631 (0.0000)		- 0.348387 (0.0003)		0.0296215 (0.6616)	0.463797 (0.00001)	0.0327799 (0.1571)	3321.458	3328.698
ARIMAX MODEL F (HNK)	- 0.292939 (0.0003)				0.0223462 (0.68480)	0.104921 (0.2112)	- 0.0173631 (0.2704)	3341.215	3347.235
ARIMAX MODEL G (HNK)			- 0.300345 (0.0003)		0.0156044 (0.7650)	0.086010 (0.2878)	- 0.0138318 (0.3640)	3341.364	3347.384
ARIMAX MODEL H (HNK)	- 0.195541 (0.5837)		- 0.106898 (0.7739)		0.0196280 (0.7176)	0.0974331 (0.2576)	- 0.0159640 (0.3232)	3343.135	3350.358

Footnote: the bold model is the best model, because it has the least AIC and BIC

Therefore, the identified ARIMAX model in Table 6 is

$$\text{Model (E): } X_t = \emptyset_1 X_{t-1} + \Theta_1 e_{t-1} + \beta_1 Y_t + \beta_2 Z_t + \beta_3 W_t + e_t \quad \text{eq.19}$$

By substitution

$$X_t = 0.98X_{t-1} - 0.35e_{t-1} + 0.03Y_t + 0.46Z_t + 0.03W_t + e_t \quad \text{eq.20}$$

Table 7: ARIMAX Models, Parameter Estimates and Model Selection Criteria for ATK

MODELS	Φ_1	Φ_2	Θ_1	Θ_2	β_1	B_2	B_3	AIC	BIC
ARIMAX MODEL A (ATK)	0.621396 (0.00001)				0.192846 (0.20302)	-0.006929 (0.83354)	0.0221819 (0.085626)	3404.604	3410.638
ARIMAX MODEL B (ATK)			0.273414 (0.00001)		-0.043570 (0.7929)	0.071058 (0.0029)	-0.166559 (0.0706)	3425.022	3431.056
ARIMAX MODEL C (ATK)	0.406174 (0.00001)	0.421911 (0.00001)			0.178386 (0.20647)	-0.024904 (0.4342)	0.0644165 (0.5214)	3380.670	3387.910
ARIMAX MODEL D (ATK)								Correlated residual	Correlated residual
ARIMAX MODEL E (ATK)	0.976141 (0.0000)		-0.491188 (0.00001)		0.166969 (0.2403)	-0.0564685 (0.0653)	0.0260587 (0.8115)	3385.839	3393.080
ARIMAX MODEL F (ATK)	-0.523016 (0.00001)				0.0734566 (0.5112)	-0.0108721 (0.4409)	-0.0134531 (0.8078)	3363.111	3369.131
ARIMAX MODEL G (ATK)			-0.549374 (0.0000)		0.113573 (0.2237)	-0.0144321 (0.1996)	-0.0339752 (0.4531)	3364.443	3370.463
ARIMAX MODEL CH (ATK)	-0.354872 (0.0108)		-0.240094 (0.0940)		0.0797568 (0.4298)	-0.0110061 (0.3807)	-0.0191156 (0.6993)	3362.318	3369.542

Footnote: the bold model is the best model, because it has the least AIC and BIC

In Table 7 the identified ARIMAX modelis

$$\text{Model F} \quad \nabla Z_t = \emptyset_1 Z_{t-1} + \beta_1 W_t + \beta_2 Y_t + \beta_3 X_t + e_t \quad \text{eq.21}$$

By substitution

$$\nabla Z_t = -0.52Z_{t-1} + 0.07W_t - 0.011Y_t - 0.013X_t + e_t \quad \text{eq.22}$$

Table 8: ARIMAX Models, Parameter Estimates and Model Selection Criteria for AGO

MODELS	Φ_1	Φ_2	Θ_1	Θ_2	β_1	B_2	B_3	AIC	BIC
ARIMAX MODEL A (AGO)	0.284591 (0.0009)	-	-	-	0.117087 (0.00001)	0.352433 (0.00001)	- 0.00327669 (0.9438)	3228.457	3234.491
ARIMAX MODEL B (AGO)			0.187145 (0.0085)		0.117545 (0.00001)	0.354126 (0.00001)	-0.0239437 (0.5840)	3232.334	3238.368
ARIMAX MODEL C (AGO)	0.224186 (0.0074)	0.266126 (0.0012)			0.119583 (0.00001)	0.334503 (0.00001)	0.0152469 (0.7545)	3220.302	3227.542
ARIMAX MODEL D (AGO)	-	-		-	-	-	-	Correlated residual	Correlated residual
ARIMAX MODEL E (AGO)	0.812209 (0.00001)		-0.553207 (0.00001)		0.123381 (0.00001)	0.312882 (0.00001)	0.0148720 (0.7553)	3220.825	3228.065
ARIMAX MODEL F (AGO)	-0.410220 (0.0000)				-0.000594 (0.9373)	0.0156424 (0.6840)	-0.009263 (0.8574)	3341.143	3347.163
ARIMAX MODEL G (AGO)			-0.440350 (0.00001)		9.55283 (0.9881)	0.0120159 (0.7097)	-0.0168409 (0.7027)	3338.246	3344.266
ARIMAX MODEL H (AGO)	-0.168750 (0.2893)		-0.310937 (0.0360)		-0.000230 (0.9716)	0.0136606 (0.6765)	-0.0146956 (0.7394)	3339.242	3346.466

Footnote: the bold model is the best model, because it has the least AIC and BIC

Similarly, in Table 8 the identified ARIMAX model is

Model C
$$W_t = \emptyset_1 W_{t-1} + \emptyset_2 W_{t-2} + \beta_1 Y_t + \beta_2 X_t + \beta_3 Z_t + e_t \quad \text{eq.23}$$

By substitution

$$W_t = 0.224W_{t-1} + 0.266W_{t-2} + 0.119Y_t + 0.335X_t + 0.015Z_t + e_t \quad \text{eq.24}$$

Identified Models Fitted Values Comparison

The identified models for both the ARIMA and ARIMAX model for each variable are obtained, then summarized in Table 9; for comparison purpose, using accuracy measures See Table9.

Table 9: Accuracy Measures Comparison between ARIMA and ARIMAX MODELS Fitted Values

MA	ARIMA MODEL				ARIMAX MODEL			
	PMS	HNK	ATK	AGO	PMS	HNK	ATK	AGO
ME	10001.31	3315.37	1818.05	160320.35	10884	2866	688.4	-945.08
MAE	85052.77	18543.06	13679.22	160320	68924	17394	14840	11658
MSE	1.652×10^{10}	7.714×10^8	8.009×10^8	3.581×10^{10}	8.640×10^9	5.525×10^8	8.914×10^8	2.777×10^8
RMSE	128536.82	27774.48	28299.66	189225.67	92950	23504	29856	16666
MAPE(%)	0.1109	0.2878	0.4106	1.0206	0.093653	0.2332	0.4238	0.1096
MAPE Percentages	11.09%	28.78%	41.06%	102.66%	9.365%	23.32%	42.38%	10.96%

Footnote: The Red Colors are the Significant Models

From Table 9, the accuracy measure show that ARIMAX models are better than ARIMA models for PMS, HNK and AGO except ATK; where the MAE, MSE and RMSE ARIMA models are smaller than ARIMAX models.

However, the identified ARIMAX models fitted values seem to performance better than the ARIMA models for three variables. Next, the forecasts values were compared.

Identified Model Forecasts values

The forecasts for both models were estimated and plot below in Table 9 and Figure13 to 15 (i.e., the year 2019 and 2020 forecast).

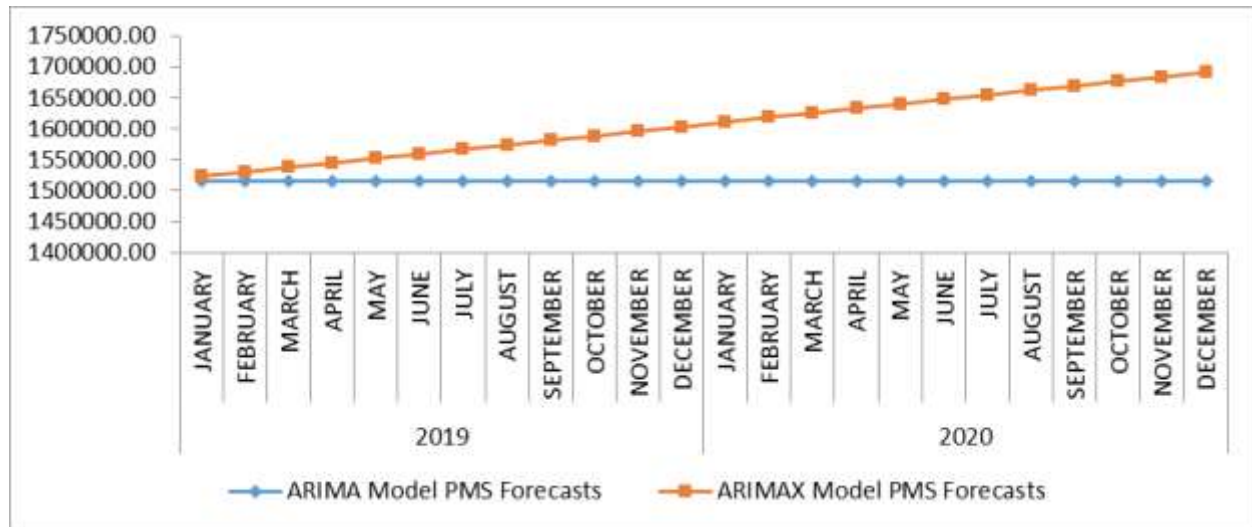


Figure 13: ARIMA and ARIMAX Model Forecasts comparison plots for PMS

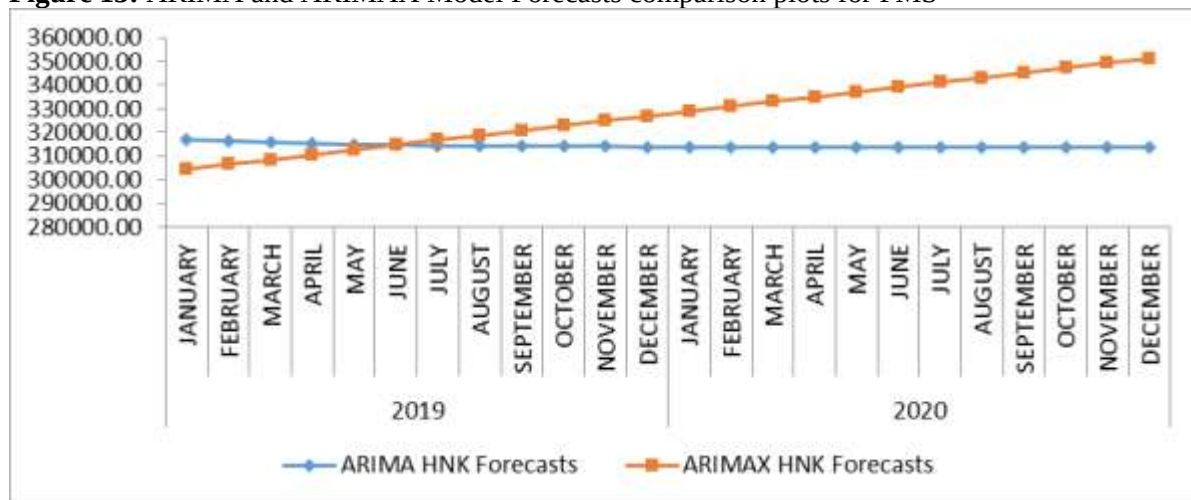


Figure 14: ARIMA and ARIMAX Model Forecasts comparison plots for HNK

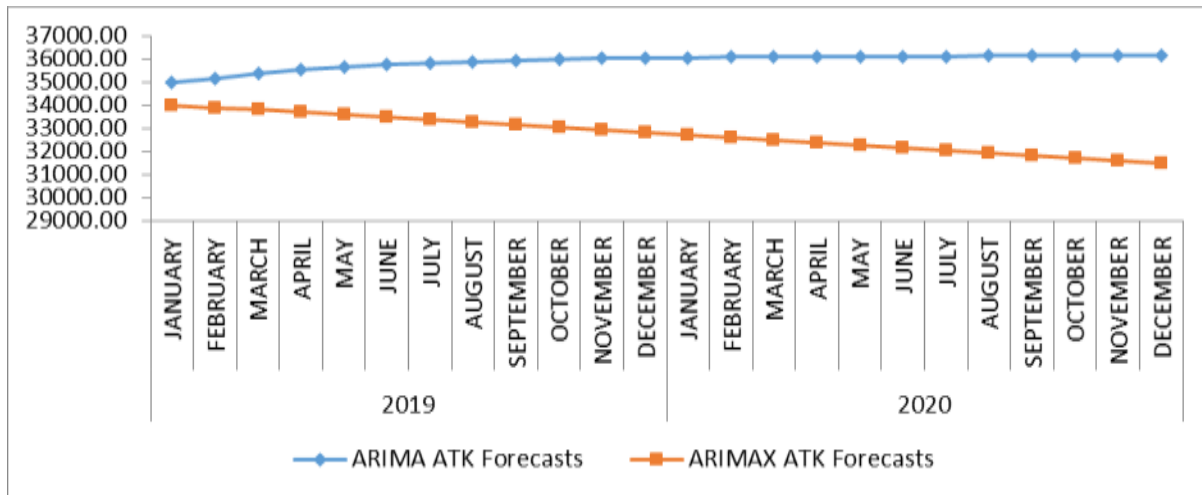


Figure 15: ARIMA and ARIMAX Model Forecasts comparison plots for ATK

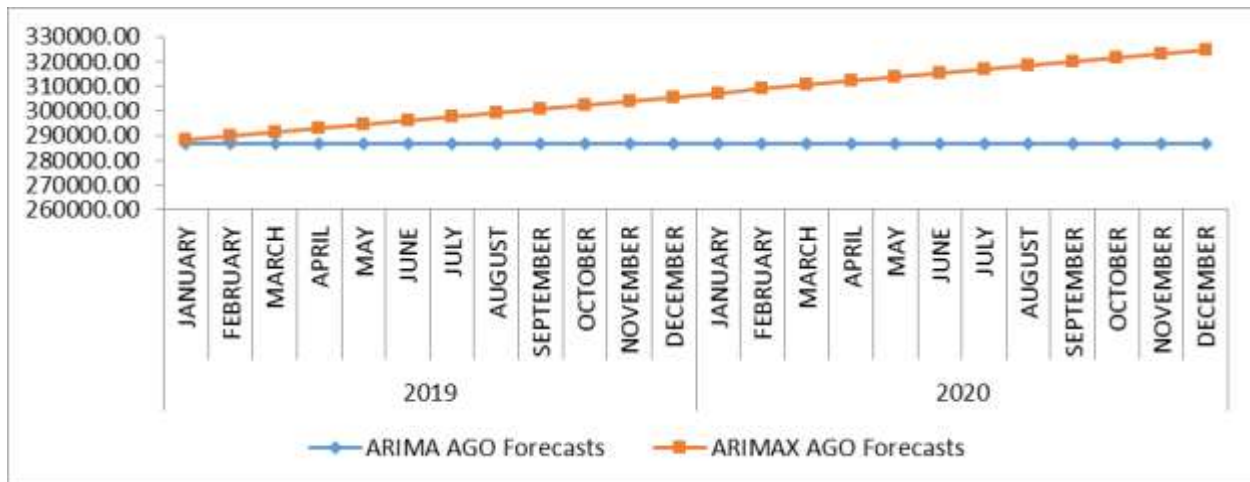


Figure 16: ARIMA and ARIMAX Model Forecasts comparison plots for AGO

The plots in Figures 4.25 to 4.28 shows that the identified ARIMAX models forecasts for 2019 and 2020 will appreciate as the years goes for three variables (PMS, HNK and AGO) except ATK variable, where the plot will depreciate. Also noted, the identified ARIMA models forecasts will only appreciate for ATK variable while it's will depreciate for PMS, HNK and AGO variables.

Hence, the identified ARIMAX models performance better than the ARIMA models for three of the variables.

5. CONCLUSION

A slightly depreciate downward/upward appreciate were noticed for the variables as the years goes. It is an indication that the series is not stationary. Then, the series were difference once to obtain stationarity (or First-difference) for the variables. In addition, the ACF plot suggest an AR(p) process while PACF suggest AR (2) since the first two lags cut. However, various ARIMA and ARIMAX models were fitted with the considered ACF and PACF were white noise. Finally, the identified ARIMAX models forecasts for 2019 and 2020 will appreciate as the years goes for three variables (PMS, HNK and AGO) except

ATK variable, where the plot will depreciate. Also noted, the identified ARIMA models forecasts only appreciate for ATK variable while it depreciates for PMS, HNK and AGO variables. Hence, the identified ARIMAX models performance better than the ARIMA models for three of the variables (PMS, HNK and AGO), using accuracy measure except ATK; where the MAE, MSE and RMSE ARIMA models are smaller than ARIMAX models. This researcher recommended ARIMAX models should be used for the forecasting of PMS, HNK and AGO for effective and reliable understanding of this variables, except ATK variable. Also, recommended ARIMA model for ATK variable. This research was able to identify ARIMAX models for PMS, HNK and AGO variables and ARIMA model for ATK variable.

REFERENCES

1. Noakes, D. J., McLeod, A. I., and Hipel, K. W. (1985).
2. Forecasting monthly river flow time series. *International Journal of Forecasting*, 1(2), 179-190. [doi:10.1016/0169-2070(85)900226]
3. Salas, J. D., Markus, M., and Tokar, A. S. (2000).
4. Stream flow forecasting based on artificial neural networks.
5. Bender, M., and Simonovic, S. (1994).
6. Time-series modeling for long-range stream-flow forecasting. *Journal of Water Resources Planning and Management*, 120(6), 857-870. [doi:10.1061/(ASCE)0733-9496(1994)120:6(857)]
7. Hipel K.W, Mcleod A.I(1994).
8. Time series modeling of water resource. 1st edition, vol. 45, ISBN: 9780080870366.
9. Abrahart, R. J., and See, L. (2000).
10. Comparing neural network and autoregressive moving average techniques for the provision of continuous river flow forecasts in two contrasting catchments. *Hydrological Processes*, 14(11-12), 2157-2172. [doi:10.1002/1099 1085(20000815/30)14:11/12<2157: AID-HYP57>3.0.CO;2-S]
11. Yyrekli K. A, Ozturk F., Testing the Residual of an ARIMA model on the cekerek stream. *Turkish J. Eng. Env. Sci.* vol. 29, p.p 61-74.
12. NNPC (2013). "Monthly crude oil export statistics," Annual statistical
13. Alsharif, M.H., Mohammad, K.Y. & Jeong, K. (2019).
14. Time series ARIMA model for prediction for daily and monthly average global solar radiation. *Symmetry* 2019, 11, 240; doi:10.3390/sym 11020240 www.mdpi.com/journal/symmetry.
15. Box, G. E. P., and Jenkins, G. M. (1976).
16. *Time Series Analysis: Forecasting and Control*. San Francisco: Holden-Day.
17. Box, G.E.P., Jenkins, G.M. and Reinsel, G.C. (2008)
18. Time Series Analysis: Forecasting and Control. 4th Edition, Wiley, Oxford. <https://doi.org/10.1002/9781118619193>
19. Box, G. E. P., Jenkins, G. M., & Reinsel, G. C. (1994).
20. *Time series analysis: forecasting and control* (3rd ed.). Prentice-Hall, Upper Saddle River, NJ.
21. Box, G.E., P. & Tiao, G.C. (1975).
22. Intervention analysis with application to economic and environmental problems. *Journal of the American statistical association*, 1975, 70, (349, (70-79).